SMART INDIA HACKATHON 2026

TITLE PAGE

Problem Statement ID – 26032
Problem Statement Title – Smart Procurement Management for Govt. Mandis
Theme – Smart Automation
PS Category – Software
Team ID – SIH2026-APP-7729
Team Name – KisanSetu AI

KisanSetu AI

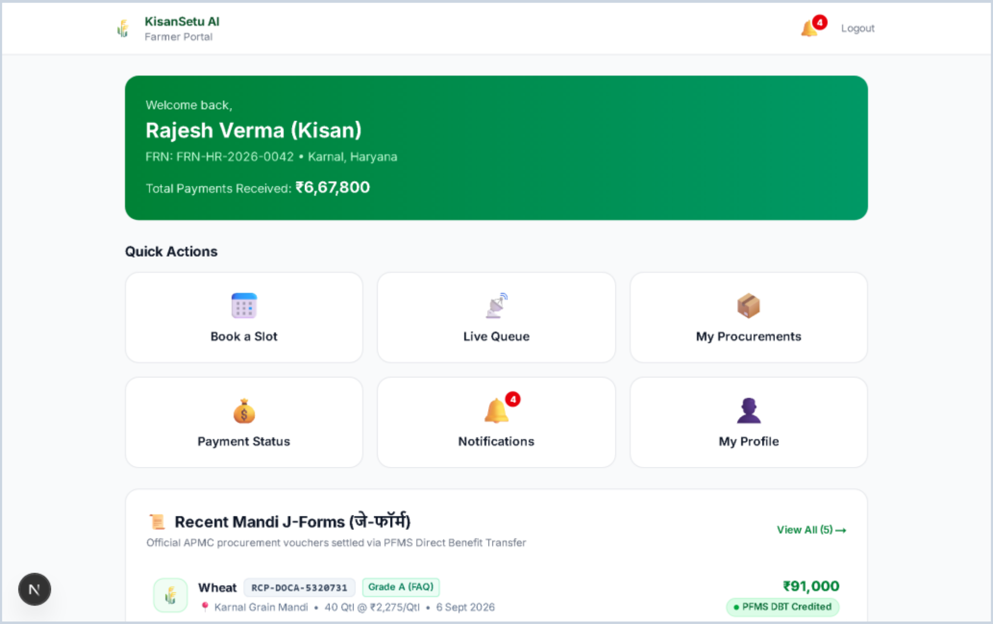
KisanSetu AI — Smart Procurement Management

SMART INDIA HACKATHON 2026

❖ Proposed Solution

- KisanSetu AI — an end-to-end smart procurement & queue management platform for Indian APMC Mandis
- Farmers book time-slots digitally from home → receive a unique digital queue token (e.g. A014)
- AI engine scores each slot for congestion (0-100) and recommends the lowest-wait option
- Real-time WebSocket queue board, Desktop Tray Notifier, & phone SMS — zero manual refresh
- Officer digitally grades crop (Grade A / Standard / Moisture %) → instant digital J-Form receipt
- Direct Benefit Transfer (DBT) tracking: PENDING → PROCESSING → COMPLETED with txn ref
- Government admin sees live national KPI dashboard — congestion heatmap, volume, payment rates

Innovation: 3-Layer AI Congestion Engine | Desktop Tray App | Multi-Channel SMS (ADB/Cloud) | Zero Paper
Live App: https://kisansetu-sih.vercel.app



🖥️ **Live System: Farmer Experience & Digital Token Workflow**

Farmers track total DBT payments (₹6.67L+ disbursed), book low-congestion slots with instant digital queue tokens (e.g. A014), and view authenticated J-Forms.

KisanSetu
AI

Technical Approach

SMART INDIA HACKATHON 2026

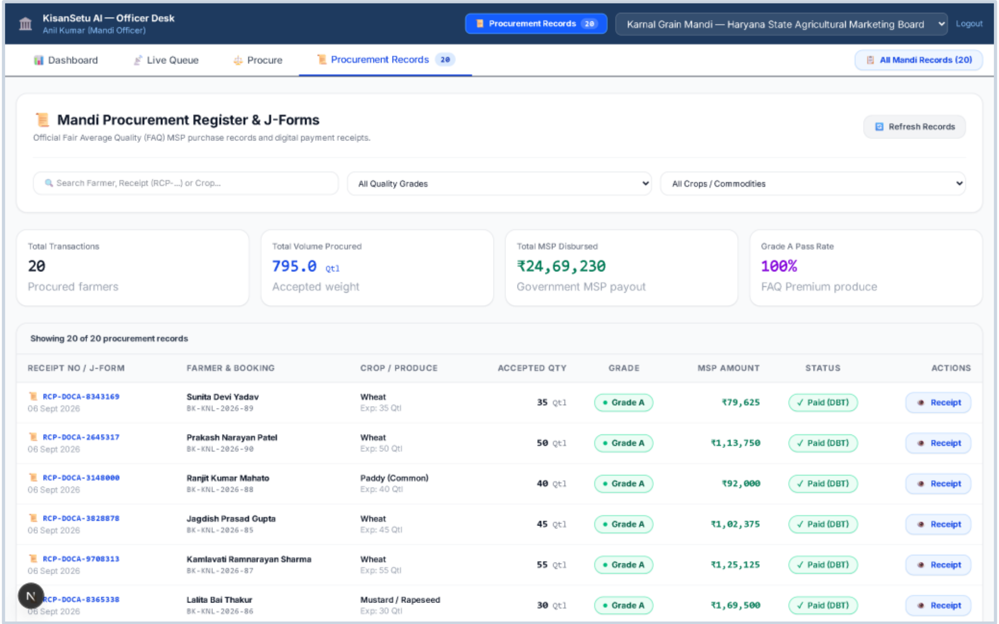
- ## Tech Stack

Frontend: Next.js 16 (TypeScript) · Tailwind CSS · App Router · WebSocket client
Backend: FastAPI (Python 3.13) · SQLAlchemy 2.0 async · Pydantic v2 · JWT auth
Database: SQLite (dev) / PostgreSQL (prod) · Supabase CLI · Zero-downtime store
Infra: Docker Compose · GitHub Actions CI/CD · Vercel · System Tray Notifier daemon
APIs: REST (12 routers) + 2 WebSocket channels + SMS Gateway (ADB / MSG91)

Methodology / Flow

Farmer → Books Slot via AI Recommendation → Receives Digital Token
Officer → Calls Token via Queue Desk / Tray Notifier → Digital Grading → J-Form
System → Instant WebSocket push + Phone SMS alert → DBT Payment PENDING
Admin → Live Heatmap updates → Monitors Mandi KPIs & Congestion in real-time

Prototype: Fully functional — 15 Mandis, 19 crops, 1200+ slots, Tray App · 13 tests passing ✅



🖥️ **Live System: Procurement Engine & Real-Time Grading**
Officer desk automates CACP MSP formulas, moisture & weighbridge deductions, generates signed digital J-Forms, and syncs queue status across mandis.

KisanSetu AI

Feasibility and Viability

- **Feasibility**

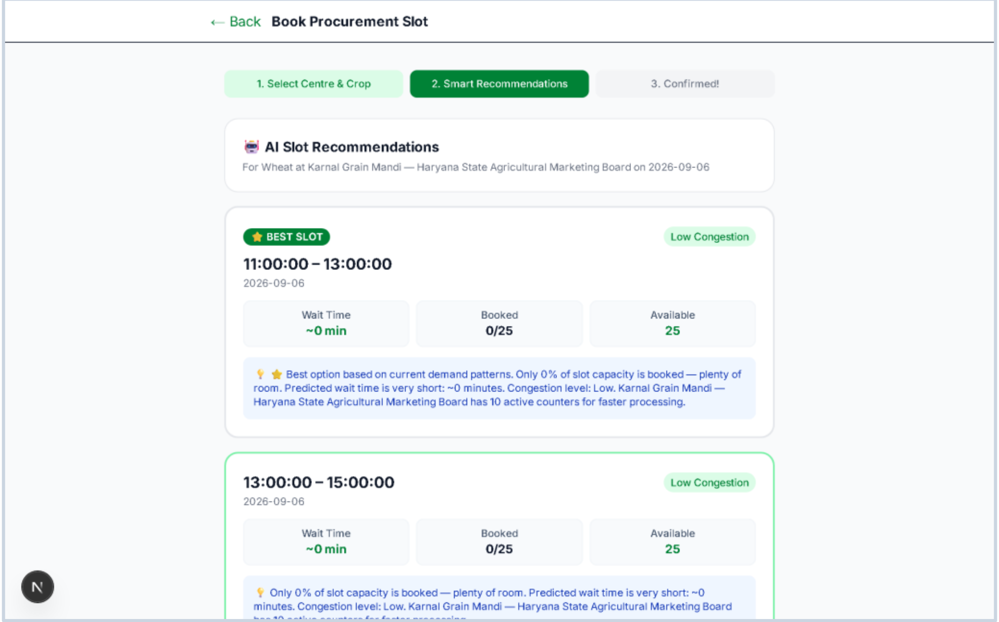
✅ Tech stack is proven, open-source, and free-tier deployable (Vercel + Supabase)
✅ One-click launcher (start.bat) boots Backend, Frontend, and Tray Notifier concurrently
✅ GitHub Actions auto-refreshes 14-day slot schedules weekly — zero cloud shutdown
✅ Scales to 7,000+ APMC Mandis via PostgreSQL read-replicas and distributed caching

Challenges & Mitigations

Challenge: Low smartphone penetration in rural farming communities
→ Solution: Multi-channel SMS engine (ADB USB bridge + MSG91 DLT gateway) reaches any 2G phone

Challenge: Internet connectivity / server downtime at remote Mandis
→ Solution: Resilient dual-layer store (local SQLite / Next.js store) ensures 100% offline uptime

Challenge: Farmer adoption & digital literacy
→ Solution: Hindi / Gujarati / regional UI; bilingual audio token calls; Mandi helper desk



🖥️ **Live System: 3-Layer AI Congestion Engine**
Real-time demand scoring recommends "⭐ BEST SLOT" with ~0 min wait times, distributing tractor traffic and preventing multi-day harvest bottlenecks.

KisanSetu
AI

Impact and Benefits

SMART INDIA HACKATHON 2026

- **Impact & Benefits**

Social Impact

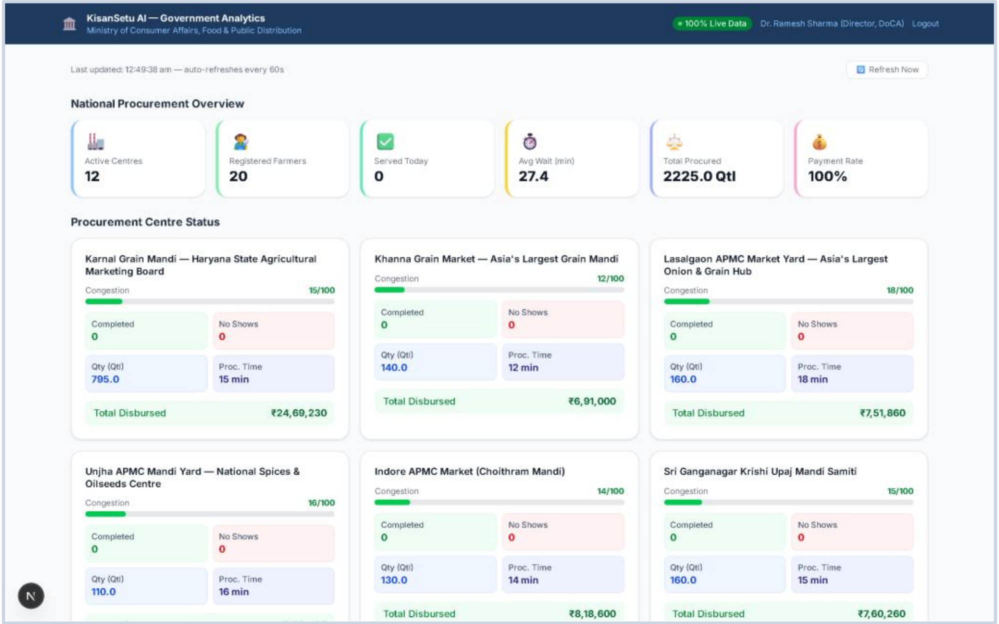
• Average mandi wait time reduced by 23.3% (simulation: 241 min → 185 min)
• Eliminates 12–48 hour tractor roadblocks at peak Rabi / Kharif harvest seasons
• Zero post-harvest produce spoilage from multi-day open-air queue bottlenecks

Economic Impact

• Tamper-proof digital J-Form receipts protect farmers against unauthorised deductions
• Direct Benefit Transfer (DBT) tracking prevents middleman & commission-agent leakage
• Target: 7,000+ APMC Mandis across India → millions of farmers directly benefited

Operational / Government Impact

• Real-time national KPI dashboard gives DoCA & Ministry officers live actionable data
• Paperless procurement audit trail — instant verification, weighbridge to bank credit



🖥️ **Live System: National Procurement Oversight & Analytics**

Live government executive dashboard monitoring 12+ APMC mandis, 2,225+ Qtl procured, 100% DBT disbursement, and real-time regional congestion indices.

KisanSetu AI

Research and References

SMART INDIA HACKATHON 2026

• Research & References

1. SIH 2026 Problem Statement 26032 — Ministry of Consumer Affairs, Food & Public Distribution
   https://www.sih.gov.in

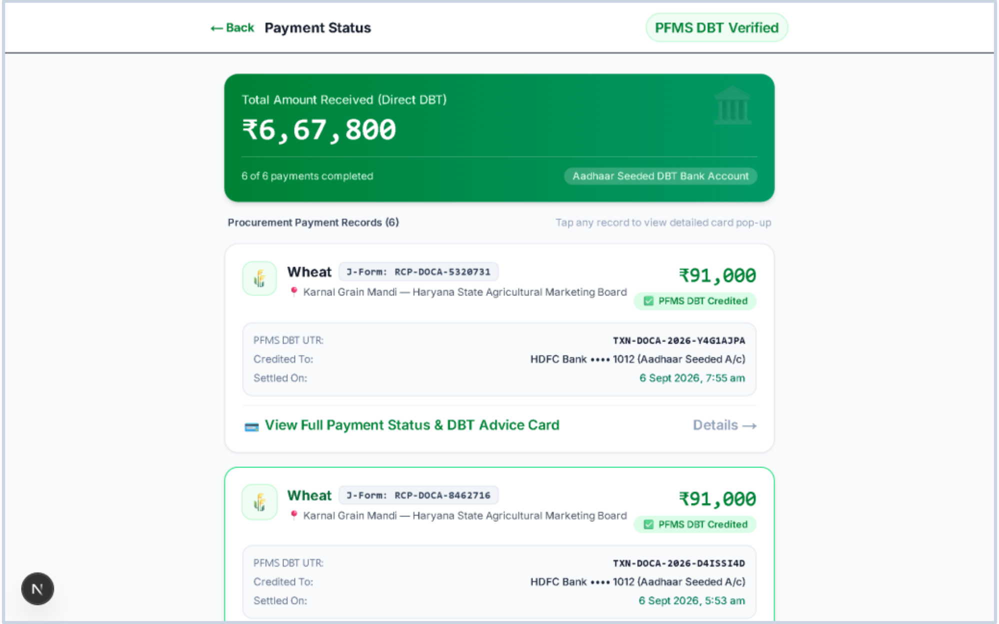
2. APMC Mandi Procurement & MSP Policy — Food Corporation of India (FCI)
   https://fci.gov.in

3. Congestion Reduction Simulation — KisanSetu AI Internal Analysis (2026)
   docs/CONGESTION_REDUCTION_ANALYSIS.md · 23.3% avg wait-time reduction validated

4. CACP Minimum Support Prices (2024–2026) — Commission for Agricultural Costs & Prices

5. Tech Stack: FastAPI 0.115 · Next.js 16 · Supabase PostgreSQL · WebSocket RFC 6455

Live App: https://kisansetu-sih.vercel.app
GitHub: https://github.com/YTxFSGAMERz/KisanSetu-AI



🖥️ **Live System: PFMS Direct Benefit Transfer (DBT) Ledger**
End-to-end transparency: verified bank credits with official PFMS UTR references directly into Aadhaar-seeded accounts, eliminating middleman leakages.